Chapter 1

Real Numbers

Fundamental Theorem of Arithmetic and Irrational Numbers

IP Important Points

A composite number can be expressed as a product of primes and this decomposition is unique, apart from the order in which prime factors occurs.

For two positive numbers a and b
HCF $(a, b) \times$ LCM $(a, b) = a \times b$.

A number which can not be expressed in the form $\frac{m}{n}$ where m and n are integers, $q \neq 0$ is called irrational number.

MCQs Multiple Choice Questions (1 Mark)

1. The ratio of HCF to LCM of the least composite number and the least prime number is: [2023]
   (a) 1 : 2 (b) 2 : 1
   (c) 1 : 1 (d) 1 : 3

2. The exponent of 5 in the prime factorisation of 3750 is [2021]
   (a) 3 (b) 4
   (c) 5 (d) 6

3. What is the greatest possible speed at which a girl can walk 95 m and 171 m in an exact number of minutes? [2021]
   (a) 17 m/min (b) 19 m/min
   (c) 23 m/min (d) 13 m/min

4. Three alarm clocks ring their alarms at regular intervals of 20 min, 25 min and 30 min respectively. If they first beep together at 12 noon, at what time will they beep again for the first time? [2021]
   (a) 4 : 00 pm (b) 4 : 30 pm
   (c) 5 : 00 pm (d) 5 : 30 pm

5. If a and b are two coprime numbers, then a^3 and b^3 are [2021]
   (a) Coprime (b) Not coprime
   (c) Even (d) Odd

6. If n is a natural number, then $2(5^n + 6^n)$ always ends with [2021]
   (a) 1 (b) 4
   (c) 3 (d) 2

7. The LCM of two numbers, is 2400. Which of the following CANNOT be their HCF? [2021]
   (a) 300 (b) 400
   (c) 500 (d) 600

8. The total number of factors of a prime number is [2020]
   (a) 1 (b) 0
   (c) 2 (d) 3

9. The HCF and the LCM of 12, 21, 15 respectively are [2020]
   (a) 3, 140 (b) 12, 420
   (c) 3, 420 (d) 420, 3

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VSA Very Short Answer Type Questions (2 Marks)

10. Two numbers are in the ratio 2 : 3 and their LCM is 180. What is the HCF of these numbers? [2023]

11. If $\sqrt{2}$ is given as an irrational number, then prove that $(5 - 2\sqrt{2})$ is an irrational number. [2023(C)]

12. Check whether 6^n can end with the digit 0 for any natural number n. [2023(C)]

13. Two positive integers a and b can be written as $a = x^3y^2$ and $b = xy^3$; x, y are prime numbers, then find the LCM(a, b). [2019]

14. Two tankers contain 850 litres and 680 litres of petrol respectively. Find the maximum capacity of a container which can measure the petrol of either tanker in exact number of times. [2016]

15. Show that $(\sqrt{3} + \sqrt{5})^2$ is an irrational number. [2015]

16. Show that 9^n cannot end with digit 0 for any natural number n. [2014]

17. Determine the values of p and q so that the prime factorisation of 2520 is expressible as $2^3 \times 3^p \times q \times 7$. [2014]

SA Short Answer Type Questions (3 Marks)

18. Prove that $\sqrt{5}$ is an irrational number. [2023]

19. Prove that $\frac{2+\sqrt{3}}{5}$ is an irrational number, given that $\sqrt{3}$ is an irrational number. [2019]

20. Find HCF and LCM of 404 and 96 and verify that HCF × LCM = Product of the two given numbers. [2018]

21. Explain whether the number $3 \times 5 \times 13 \times 46 + 23$ is a prime number or a composite number. [2016]

LA Long Answer Type Question (5 Marks)

22. Find HCF of 378, 180 and 420 by prime factorisation method. Is HCF × LCM of three numbers equal to the product of the three numbers? [2014]

CBQs Case-Based Question

23. February 14 is celebrated as International Book Giving Day and many countries in this world celebrate this day. Some people in India also started celebrating this day and donated the following number of books of various subjects to a public library:

History = 96, Science = 240, Mathematics = 336

These books have to be arranged in minimum number of stacks such that each stack contains books of only one subject and the number of books on each stack is the same.

Based on the above information, answer the following questions: [2023(C)]

(*i*) How many books are arranged in each stack?

(*ii*) How many stacks are used to arrange all the Mathematics books?

(*iii*) Determine the total number of stacks that will be used for arranging all the books.

OR

If the thickness of each book of History, Science and Mathematics is 1.8 cm, 2.2 cm and 2.5 cm respectively, then find the height of each stack of History, Science and Mathematics books.

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Very short Answer Type Questions

1. Prove that $\sqrt{2}$ is irrational.
2. Prove that $\sqrt{6}$ is irrational.
3. Find the greatest possible 3-digit number that leaves a remainder of 3 when divided by 5.
4. Find the smallest positive number which when divided by 35, 56 and 91 leave the same remainder 7 in each case.
5. Find the HCF and LCM of 6, 72 and 120 using prime factorisation method.
6. Check whether the statement below is true or false. "The square root of every composite number is rational." Justify your answer by proving rationality or irrationality as applicable.
7. There is a circular path around a sports field. Sonia takes 18 minutes to drive one round of the field, while Ravi takes 12 minutes for the same. Suppose they both start at the same point and at the same time, and go in the same direction. After how many minutes will they meet again at the starting point?
8. Write one rational and one irrational number lying between 0.25 and 0.32.
9. Show that every positive odd integer is of the form $(4q+1)$ or $(4q+3)$ for some integer q.
10. Check whether 531111 and $5 \times 7 \times 7 \times 3+3$ are composite numbers and justify your answer.
11. Find the prime factorisation of the denominator of rational number expressed as 6.12 in simplest form.
12. The HCF of two numbers is 23 and their LCM is 1449. If one of the numbers is 161, find the other number.

Short Answer Type Questions

1. Grow More Plantation have two rectangular fields of the same width but different lengths they are required to plant 84 trees in the smaller field and 231 trees in the larger field. In both fields the trees will be planted in the same number of rows but in different number of columns.
   i. What is the greatest number of rows that can be planted in this arrangement, show your work.
   ii. If the trees will be planted in the number of rows obtained in part
   (i) then how many columns each field will have?
2. If p is a prime number, then show that $\sqrt{p}$ is irrational number.
3. Given that $\sqrt{3}$ is irrational prove that $5+2\sqrt{3}$ is irrational.
4. Show that $3\sqrt{5}$ is irrational.

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5. The LCM of 64, 82 and k is 124 where k is a positive integer. Find the smallest value of k. Show your steps.

6. Consider the numbers 4^n, where n is a natural number. Check whether there is any value of n for which 4^n ends with the digit zero.

7. Prime factorisation of three numbers A, B and C is given below:
$A = (2^r \times 3^p \times 5^q)$
$B = (2^p \times 3^r \times 5^p)$
$C = (2^q \times 3^q \times 5^p)$ such that,
$p < q < r$ and p, q, and r are natural numbers.
   - The largest number that divides A, B and C without leaving a remainder is 30.
   - The smallest number that leaves a remainder of 2 when divided by each of A, B and C is 5402.

   Find A, B and C.
   Show your work.

8. Find the LCM. and HCF of 26 and 91 and verify that HCF × LCM = product of two numbers.

9. Prove that $(3 - \sqrt{5})$ is irrational, given that $\sqrt{5}$ is irrational.

10. Prove that the difference and the quotient of $(3 + 2\sqrt{3})$ and $(3 - 2\sqrt{3})$ are irrational.

11. There are 144 cartons of Coke cans and 90 cartons of Pepsi cans that need to be stacked. If each stack is of the same height and is to contain cartons of the same drink, what would be the greatest number of cartons each stack would have?

12. Using prime factorisation method, find the HCF and LCM of 30, 72 and 432. Also show that HCF × LCM ≠ Product of the three numbers.

13. Karan has 180 blue marbles and 150 red marbles. He wants to pack them into packets containing equal number of marbles of the same colour. What is the maximum number of marbles that each packet can hold?

14. Show that every positive even integer is of the form 2q and every positive odd integer is of the form 2q + 1 for some integer q.

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MOCK TEST

MCQ Type Questions

1. Which of these is a RATIONAL number?
   a) 3π
   b) $5\sqrt{5}$
   c) 0.3466666...
   d) 0.345210651372849...

2. In a formula racing competition, the time taken by two racing cars A and B to complete 1 round of the track is 30 minutes and p minutes respectively. If the cars meet again at the starting point for the first time after 90 minutes and the HCF (30, p) = 15, then the value of p is:
   a) 45 minutes b) 60 minutes
   c) 75 minutes d) 180 minutes

3. A number q is prime factorised as $3^2 \times 7^2 \times b$, where b is a prime number other than 3 and 7.
   Based on the above information, two statements are given below - one labelled Assertion (A) and the other labelled Reason (R). Read the statements carefully and choose the option that correctly describes statements (A) and (R).
   Assertion (A): q is definitely an odd number.
   Reason (R): $3^2 \times 7^2$ is an odd number.
   a) Both (A) and (R) are true and (R) is the correct explanation for (A).
   b) Both (A) and (R) are true but (R) is no the correct explanation for (A).
   c) (A) is true but (R) is false.
   d) (A) is false but (R) is true.

4. Two statements are given below - one labelle Assertion (A) and the other labelled Reason (R). Read the statements carefully and choos the option that correctly describes statemen (A) and (R).
   Assertion (A): 2 is a prime number.
   Reason (R): The square of an irrational number is always a prime number.
   a) Both (A) and (R) are true and (R) is th correct explanation for (A).
   b) Both (A) and (R) are true but (R) is n the correct explanation for (A).
   c) (A) is true but (R) is false.
   d) (A) is false but (R) is true.

5. LCM of 6 and 20 is:
   a) 2 b) 60
   c) 120 d) 30

Very Short Answer Type Questions

6. The fraction $\frac{7}{24}$ has a terminating or non- terminating decimal expansion? Explain your answer.

7. Check whether the statement below is true or false.
   "The product of two irrational numbers is always an irrational number."
   Justify your answer by proving rationality irrationality as applicable.

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8. A forester wants to plant 66 apple trees, 88 banana trees and 110 mango trees in equal rows (in terms of number of trees). Also, he wants to make distinct roots of the trees (only one type of tree in one row). Find the minimum number of rows required.

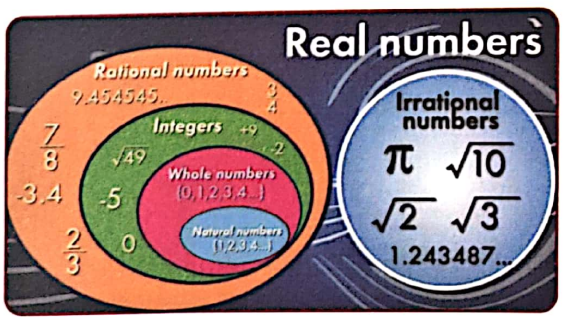
Short Answer Type Questions

11. P and Q are two positive integers such that $P = p^3$ and $Q = (pq)^2$, where p and q are prime numbers. What is the LCM (P, Q)?

12. If length breadth and height of a room is 825 cm, 675 cm and 450 cm, then find the length of the longest rod that can be used to measure the three dimensions of the room exactly.

Case Study Based Questions

16. In a class teacher is explaining real number system to make the student understand about real number: Real numbers are the numbers which include both rational and irrational numbers. Rational numbers such as integers (…-1, 0, 1 ,2…), fractions ($\frac{7}{24}$, 2.5 etc.), and irrational numbers ($\sqrt{2}$, π etc) are all real numbers.



Based on the above information answer the following questions.

i) Which of the following is not irrational number.

a) $\sqrt{2}$ b) $\frac{432}{60}$

c) $\frac{1}{2}$ d) 4π

ii) Write the two integers between -2 and +2.

iii) The product of a rational and an irrational number is rational or irrational? give one example.

9. Find the smallest pair of 4-digit numbers such that the difference between them is 303 and their HCF is 101. Show your steps.

10. Show that $6+3\sqrt{5}$ is an irrational number. Given that $\sqrt{5}$ is an irrational number.

13. Prove that $5+6\sqrt{7}$ is irrational.

14. The LCM of two numbers is 45 times of their HCF if one of the numbers is 125 and sum of their LCM and HCF is 1150, find the other number.

15. Find the greatest 6-digit number which is exactly divisible by 15, 24 and 36, show your steps.

17. A transportation survey group is observing the three traffic lights at three different road crossings change after every 48 seconds, 72 seconds and 108 seconds respectively.



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"Based on the above information, answer the following questions.

i. What is the prime factorization of 108.

ii. Find the HCF of (48, 72, 108).

iii. If the three traffic lights change simultaneously at 10 a.m., at what time will they change simultaneously again?

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